

Regular expression:

1. The language of all words: $(a+b)^*$

Valid Strings:

1. "" (Empty string - allowed since $*$ allows zero occurrences)
2. "abba" (Only contains a and b)
3. "bbbbaa" (Only contains a and b)

Invalid Strings:

1. "abc" (Contains c, which is not part of $\{a, b\}$)
2. "123" (Contains digits, which are not part of $\{a, b\}$)
3. "a_b" (Contains $_$, which is not allowed)

2. All words ending with b: $(a+b)^*b$

Valid Strings:

1. "b" (Ends with b)
2. "aab" (Ends with b)
3. "abbaabb" (Ends with b)

Invalid Strings:

1. "" (Empty string does not end with b)
2. "aaa" (Ends with a, not b)
3. "ababab" (Ends with b, but contains non a or b characters)

3. All words that start with a: $a(a+b)^*$

Valid Strings:

1. "a" (Starts with a)
2. "ababa" (Starts with a)
3. "aabb" (Starts with a)

Invalid Strings:

1. "" (Empty string does not start with a)
2. "baba" (Starts with b, not a)
3. "bb" (Starts with b, not a)

4. The language of all strings, not beginning with b: $a(a+b)^* + \Lambda$

Valid Strings:

1. "" (Empty string is explicitly allowed)
2. "a" (Starts with a)
3. "aabb" (Starts with a)

Invalid Strings:

1. "b" (Starts with b)
2. "bb" (Starts with b)
3. "bab" (Starts with b)

5. All words that start with a double letter: $(aa+bb)(a+b)^*$

Valid Strings:

1. "aa" (Starts with aa)
2. "bba" (Starts with bb)
3. "aabbab" (Starts with aa)

Invalid Strings:

1. "a" (Too short, does not contain aa or bb)
2. "ba" (Starts with b, not bb)
3. "abba" (Starts with ab, not aa or bb)

6. All words that contain at least one double letter: $(a+b)^*(aa+bb)(a+b)^*$

Valid Strings:

1. "aa" (Contains aa)
2. "bbaa" (Contains bb)
3. "ababbbaa" (Contains bb)

Invalid Strings:

1. "a" (No aa or bb)
2. "bababa" (No aa or bb)
3. "" (No aa or bb)

7. All words that contain at least two as or two bs: $b^*ab^*ab^*+a^*ba^*ba^*$

Valid Strings:

1. "aa" (Contains at least two as)
2. "bab" (Contains at least two bs)
3. "abba" (Contains both aa and bb)

Invalid Strings:

1. "a" (Contains only one a)
2. "b" (Contains only one b)
3. "" (Empty string does not contain aa or bb)

8. All words that start and end with a double letter: $(aa+bb)(a+b)^*(aa+bb)$

Valid Strings:

1. "aabb" (Starts with aa, ends with bb)
2. "bbaa" (Starts with bb, ends with aa)
3. "aabbbaa" (Starts with aa, ends with aa)

Invalid Strings:

1. "a" (Too short, does not contain double letters)
2. "abba" (Starts with a, not aa or bb)
3. "bbab" (Starts with bb but does not end with aa or bb)

9. All words of length ≥ 3 : $(a+b)(a+b)(a+b)(a+b)^*$

Valid Strings:

1. "aaa" (Length 3)
2. "bab" (Length 3)
3. "bbbb" (Length 4)

Invalid Strings:

1. "" (Empty string, length 0)
2. "a" (Length 1)
3. "ab" (Length 2)

10. All words that contain exactly one a or exactly one b: $b^*ab^*+a^*ba^*$

Valid Strings:

1. "a" (Exactly one a)
2. "bbabb" (Exactly one a)
3. "abbbb" (Exactly one a)

Invalid Strings:

1. "aa" (Contains two as)
2. "bb" (No a or a single b)
3. "aba" (Contains two as)

11. All words that don't end in "ba": $(a+b)^*(aa+ab+bb)$

Valid Strings:

1. "aa" (Ends in "aa")
2. "abab" (Ends in "ab")
3. "bbb" (Ends in "bb")

Invalid Strings:

1. "ba" (Ends in "ba")
2. "abba" (Ends in "ba")
3. "bbba" (Ends in "ba")

12. All strings of as and bs in which either the strings are all bs or there is at least one a followed by some bs: b^+ab^*

Valid Strings:

1. "bbbb" (All bs)
2. "ab" (Contains one a followed by bs)
3. "abb" (Contains one a followed by bs)

Invalid Strings:

1. "a" (No b after a)
2. "ba" (a is not followed by bs)
3. "aba" (Contains an a not followed only by bs)

13. Language of all words that have at least two as: $(a+b)^*a(a+b)^*a(a+b)^*$

Valid Strings:

1. "aa" (Contains two as)
2. "babaa" (Contains two as)
3. "abbaaba" (Contains more than two as)

Invalid Strings:

1. "a" (Only one a)
2. "bbbb" (No as at all)
3. "" (Empty string)

14. Language of all words that have at least one a and at least one $a(a+b)^*b(a+b)^+ (a+b)^*b(a+b)^*a(a+b)^*$

Valid Strings:

1. "ab" (Contains both a and b)
2. "ba" (Contains both a and b)

3. "babb" (Contains a and b)

Invalid Strings:

1. "aaaa" (Only as)
2. "bbbb" (Only bs)
3. "" (Empty string)

15. Language of all words that have at least one a or at least one b

$(a+b)^*a(a+b)^*+(a+b)^*b(a+b)^*$

Valid Strings:

1. "a" (Contains a)
2. "b" (Contains b)
3. "abba" (Contains both a and b)

Invalid Strings:

1. "" (Empty string)
2. "#" (Not a or b)
3. "123" (Not a or b)

16. The languages L , of even length, defined over $\Sigma = \{a, b\}$ $((a+b)(a+b))^*$

Valid Strings:

1. "ab" (Length 2)
2. "aabb" (Length 4)
3. "bbaa" (Length 4)

Invalid Strings:

1. "a" (Length 1, odd)
2. "aaa" (Length 3, odd)
3. "baaba" (Length 5, odd)

17. The languages L , of odd length, defined over $\Sigma = \{a, b\}$ $((a+b)(a+b))^*(a+b)$

Valid Strings:

1. "a" (Length 1)
2. "bab" (Length 3)
3. "abbab" (Length 5)

Invalid Strings:

1. "" (Empty string, not odd)
2. "bb" (Length 2, even)
3. "aabb" (Length 4, even)

18. The strings of length 2, starting with a aa+ab

Valid Strings:

1. "aa" (Starts with a, length 2)
2. "ab" (Starts with a, length 2)

Invalid Strings:

1. "b" (Too short)
2. "ba" (Starts with b)
3. "aaa" (Too long)

19. EVEN-EVEN language (aa+bb+(ab+ba)(aa+bb)*(ab+ba))

Valid Strings:

1. "aa" (Even-length block)
2. "bbaa" (Valid even sequence)
3. "abba" (Valid even sequence)

Invalid Strings:

1. "a" (Odd-length)
2. "abb" (Invalid structure)
3. "ababaa" (Contains an odd-length part)

20. ODD-ODD Language (a+b)((a+b)(a+b))* + (ab+ba)((ab+ba)(ab+ba)*)

Valid Strings:

1. "a" (Length 1, odd)
2. "bab" (Length 3, odd)
3. "abbab" (Length 5, odd)

Invalid Strings:

1. "" (Empty string, not odd)
2. "bb" (Length 2, even)
3. "aabb" (Length 4, even)

21. Set of strings with at least two letters, starting and ending with a with only bs inside ab*a

Valid Strings:

1. "aa" (Valid, no bs inside)
2. "aba" (Valid, b inside)
3. "abbbba" (Valid, multiple bs inside)

Invalid Strings:

1. "a" (Too short, needs at least two letters)
2. "baba" (Contains an extra a in the middle)
3. "abbab" (Does not end with a)

22. Language of all strings where all as come before all bs a^*b^* **Valid Strings:**

1. "a" (Only as)
2. "b" (Only bs)
3. "aaaabb" (All as before all bs)

Invalid Strings:

1. "ba" (b comes before a)
2. "abbab" (bs and as are mixed)
3. "bab" (b comes before a)

23. Language of odd-length strings over $\{a\}$ $a(aa)^*$ or $(aa)^*a$ **Valid Strings:**

1. "a" (Length 1, odd)
2. "aaa" (Length 3, odd)
3. "aaaaa" (Length 5, odd)

Invalid Strings:

1. "" (Empty string, length 0)
2. "aa" (Length 2, even)
3. "aaaa" (Length 4, even)

24. Language of even-length strings over $\{a\}$ $(aa)^*$ **Valid Strings:**

1. "" (Length 0, even)
2. "aa" (Length 2, even)
3. "aaaa" (Length 4, even)

Invalid Strings:

1. "a" (Length 1, odd)
2. "aaa" (Length 3, odd)
3. "aaaaa" (Length 5, odd)

25. Strings with no double letters (aa or bb are not allowed) $b(ab)^*a + a(ba)^*b$

Valid Strings:

1. "ab" (Alternates)
2. "abab" (Alternates)
3. "bababa" (Alternates)

Invalid Strings:

1. "aa" (Contains aa)
2. "bb" (Contains bb)
3. "abba" (Contains bb)

26. Language of strings of length exactly 3 $(a+b)^3$ or $(a+b)(a+b)(a+b)$

Valid Strings:

1. "aaa" (Length 3)
2. "abb" (Length 3)
3. "bab" (Length 3)

Invalid Strings:

1. "aa" (Length 2, too short)
2. "aaaa" (Length 4, too long)
3. "" (Empty string)

27. Language of words that have at least two as

$(a+b)^*a(a+b)^*a(a+b)^*$

Valid Strings:

1. "aa" (Exactly two as)
2. "baba" (Contains two as)
3. "abbbbaa" (Contains two as)

Invalid Strings:

1. "a" (Only one a)
2. "bbbb" (No as at all)
3. "" (Empty string)

28. Language of words with exactly two as $b^*ab^*ab^*$

Valid Strings:

1. "aa" (Two as)
2. "bab" (Exactly two as)
3. "babb" (Exactly two as)

Invalid Strings:

1. "aaa" (More than two as)
2. "bbbb" (No as)
3. "a" (Only one a)

29. Language of words with exactly two bs or exactly two as $b^*ab^*ab^*+a^*ba^*ba^*$

Valid Strings:

1. "aab" (Exactly two as)
2. "bba" (Exactly two bs)
3. "abba" (Exactly two bs)

Invalid Strings:

1. "aaa" (More than two as)
2. "bbbb" (More than two bs)
3. "abab" (More than two of both letters)

30. Language of words that contain both a and b $(a + b)^*a(a + b)^*b(a + b)^* + bb^*aa^*$

Valid Strings:

1. "ab" (Contains a and b)
2. "ba" (Contains a and b)
3. "aabb" (Contains a and b)

Invalid Strings:

1. "aaaa" (Only as)
2. "bbbb" (Only bs)
3. "" (Empty string)

31. Set of all strings that contain a double letter at some point

$$(a+b)^*(aa+bb)(a+b)^*$$

Valid Strings:

1. "aa" (Has aa)
2. "abba" (Has bb)
3. "bbaa" (Has bb and aa)

Invalid Strings:

1. "ab" (No double letters)
2. "bab" (No double letters)
3. "ababab" (No double letters)

32. Language of even number of as, followed by an odd number of bs or vice versa

$$(aa)^*b(bb)^* + (bb)^*a(aa)^*$$

Valid Strings:

1. "aabb" (Even as, odd bs)
2. "bbaaa" (Even bs, odd as)
3. "aaaaabb" (Even as, odd bs)

Invalid Strings:

1. "aab" (Even as but even bs)
2. "baa" (Odd bs, odd as)
3. "bbbbbb" (Even bs, no as)

33. Language of words without a double a

$$b^*(abb^*)a$$

Valid Strings:

1. "ab" (No aa)
2. "baba" (No aa)
3. "bababa" (No aa)

Invalid Strings:

1. "aa" (Has aa)
2. "bbaa" (Has aa)
3. "aabba" (Has aa)

34. The Language of Words with at Least Two as

$$b^*ab^*ab^*$$

Valid Strings:

1. "aa" (Contains two as)
2. "baba" (Contains two as)
3. "abbbbaa" (Contains two as)

Invalid Strings:

1. "a" (Only one a)
2. "bbbb" (No as at all)
3. "" (Empty string)

35. The Language of Words that Do Not Contain Both a and b

$(a+b)^*a(a+b)^*b(a+b)^* + bb^*aa^*+a^*+b^*$

Valid Strings (Only as, only bs, or empty):

1. "aaa" (Only as)
2. "bbbb" (Only bs)
3. "" (Empty string)

Invalid Strings (Contain both a and b):

1. "ab" (Contains both a and b)
2. "baaa" (Contains both a and b)
3. "babab" (Contains both a and b)

36. Strings of All Words Having Exactly Two as

$b^*ab^*ab^*$

Valid Strings:

1. "aa" (Two as, no bs)
2. "bab" (Two as, one b)
3. "babb" (Two as, multiple bs)

Invalid Strings:

1. "aaa" (More than two as)
2. "bbbb" (No as)
3. "a" (Only one a)

37. Language of Words Starting with a and Ending with b, or Starting with b and Ending with aa(a+b)*b+b(a+b)*a**Valid Strings:**

1. "ab" (Starts with a, ends with b)
2. "bab" (Starts with b, ends with a)
3. "abb" (Starts with a, ends with b)

Invalid Strings:

1. "aa" (Does not end with b)
2. "bb" (Does not start and end with different letters)
3. "baab" (Starts with b, but does not end with a)

38. Strings of words Where length(x) < 3, Starting with "ba"

ba

Valid Strings:

1. "ba"

Invalid Strings:

1. "b" (Too short)
2. "bab" (Too long)
3. "aa" (Does not start with "ba")

39. Strings Starting with Double b and Ending with Either a or b

bb(a+b)

Valid Strings:

1. "bba" (Starts with bb, ends with a)
2. "bbb" (Starts with bb, ends with b)
3. "bbabb" (Starts with bb, ends with b)

Invalid Strings:

1. "ba" (Does not start with bb)
2. "b" (Too short, missing second b)
3. "aaa" (Does not start with bb)

40. All Strings Ending in aa or bb

$(a+b)^*(aa+bb)$ or $(a+b)^*aa + (a+b)^*bb$

Valid Strings:

1. "aa" (Ends in aa)
2. "bbaa" (Ends in aa)
3. "abb" (Ends in bb)

Invalid Strings:

1. "a" (Does not end in aa or bb)
2. "ab" (Does not end in aa or bb)
3. "ba" (Does not end in aa or bb)

41. Any String That Contains at Least Two Consecutive as $(a+b)^*aa(a+b)^*$ **Valid Strings:**

1. "aa" (Contains "aa")
2. "baab" (Contains "aa")
3. "abaab" (Contains "aa")

Invalid Strings:

1. "a" (Does not contain "aa")
2. "babab" (No "aa" present)
3. "bb" (No "aa" present)

42. All Words That Don't End with "ba" $(a+b)^*(aa+ab+bb)$ **Valid Strings:**

1. "aa" (Ends with "aa")
2. "babab" (Ends with "ab")
3. "bbbb" (Ends with "bb")

Invalid Strings:

1. "ba" (Ends with "ba")
2. "abba" (Ends with "ba")
3. "bbbba" (Ends with "ba")

43. All Words That Start with a Double Letter $(aa+bb)(a+b)^*$ **Valid Strings:**

1. "aa" (Starts with "aa")
2. "bbabb" (Starts with "bb")
3. "aab" (Starts with "aa")

Invalid Strings:

1. "ab" (Does not start with "aa" or "bb")
2. "ba" (Does not start with "aa" or "bb")

3. "b" (Too short, missing second character)

44. All Words That Contain at Least One Double Letter $(a+b)^*(aa+bb)(a+b)^*$

Valid Strings:

1. "aa" (Contains "aa")
2. "abba" (Contains "bb")
3. "bbaaa" (Contains "bb" and "aa")

Invalid Strings:

1. "ab" (No "aa" or "bb")
2. "ba" (No "aa" or "bb")
3. "abab" (No "aa" or "bb")

45. All Words That Start and End with a Different Double Letter

$$aa(a+b)^*bb(a+b)^* + bb(a+b)^*aa(a+b)^*$$

Valid Strings:

1. "aabb" (Starts "aa", ends "bb")
2. "bbaaaa" (Starts "bb", ends "aa")
3. "aabbbb" (Starts "aa", ends "bb")

Invalid Strings:

1. "aa" (Does not end with "bb")
2. "bb" (Does not end with "aa")
3. "abab" (Does not start and end with different double letters)

46. All Words of Length > 3 $(a+b)(a+b)(a+b)(a+b)^*$ or $(a+b)(a+b)(a+b)(a+b)^+$

Valid Strings:

1. "aaaa" (Length = 4)
2. "babaab" (Length = 6)
3. "abbab" (Length = 5)

Invalid Strings:

1. "aaa" (Length = 3, does not satisfy >3)
2. "bb" (Length = 2)
3. "aba" (Length = 3)

47. All Words of Length ≥ 3 $(a+b)(a+b)(a+b)(a+b)^*$ or $(a+b)(a+b)(a+b)^+$

Valid Strings:

1. "aaa" (Length = 3)
2. "baba" (Length = 4)
3. "abbab" (Length = 5)

Invalid Strings:

1. "aa" (Length = 2)
2. "b" (Length = 1)
3. "" (Empty string)

48. All Words That Start and End with a Double Letter

$aa(a+b)^*bb + bb(a+b)^*aa$ or $(aa+bb)(a+b)^*(aa+bb)$

Valid Strings:

1. "aabb" (Starts "aa", ends "bb")
2. "bbabba" (Starts "bb", ends "aa")
3. "aabbaab" (Starts "aa", ends "bb")

Invalid Strings:

1. "aa" (Ends incorrectly, no "bb")
2. "bb" (Ends incorrectly, no "aa")
3. "abba" (Does not start and end with different double letters)

49. All Words That Contain Exactly One a or One b $b^*ab^*a^*ba^*$

Valid Strings:

1. "a" (Exactly one a)
2. "b" (Exactly one b)
3. "ba" (One b, one a)

Invalid Strings:

1. "aa" (More than one a)
2. "bbb" (No a present)
3. "abab" (Contains multiple as and bs)

50. All Words That Contain Exactly Two as or Exactly Two bs

$(a+b)^*aa(a+b)^* + (a+b)^*bb(a+b)^*$

Valid Strings:

1. "aabb" (Exactly two as)
2. "bbaaa" (Exactly two bs)

3. "aba" (Exactly two as)

Invalid Strings:

1. "aaa" (More than two as)
2. "bbb" (More than two bs)
3. "abba" (More than two of both)

51. The Language of Strings in Which Any Number of as May Occur Before, Between, and After the bs $a^*ba^*ba^*$

Valid Strings:

1. "bbaa" (Two bs, any number of as around them)
2. "abba" (Two bs with as around them)
3. "aaaaabbbaaaaa" (Two bs, any number of as)

Invalid Strings:

1. "a" (No bs)
2. "b" (Only one b, must have at least two)
3. "bb" (No as present, but some as must occur)

52. All Words That Contain at Least Two as $b^*ab^*ab^*$

Valid Strings:

1. "aab" (Contains two as)
2. "baab" (Contains two as)
3. "bbabb" (Contains two as)

Invalid Strings:

1. "a" (Only one a)
2. "bbbb" (No as)
3. "ba" (Only one a)

53. All Words With at Least Two as $b^*ab^*a(a+b)^*$

Valid Strings:

1. "aab" (Two as, valid)
2. "baabb" (Two as, valid)
3. "abbaba" (Contains two as)

Invalid Strings:

1. "a" (Only one a)

2. "b" (No as)
3. "bbbb" (No as)

54. All Words With Exactly Two as $b^*ab^*ab^*b^*ab^*ab^*b^*ab^*ab^*$

Valid Strings:

1. "aab" (Exactly two as)
2. "bab" (Exactly two as)
3. "bbaabb" (Exactly two as)

Invalid Strings:

1. "aaa" (More than two as)
2. "bb" (No as)
3. "ababa" (More than two as)

55. Language That Defines All Even-Length Strings of Alternating as and bs

$(ab)^*(ba)^*$

Valid Strings:

1. "ab" (Even length, alternating)
2. "ba" (Even length, alternating)
3. "abab" (Even length, alternating)

Invalid Strings:

1. "a" (Odd length)
2. "b" (Odd length)
3. "abb" (Not alternating)

56. Language That Defines All Odd-Length Strings of Alternating as and bs

$(a+bb+ab+ba)^*$

Valid Strings:

1. "a" (Odd length, allowed)
2. "bab" (Odd length, alternating)
3. "abba" (Odd length, alternating)

Invalid Strings:

1. "aa" (Even length)
2. "bb" (Even length)
3. "ababa" (Even length)

57. The Set of Strings With an Even Number of as Followed by an Odd Number of bs

$(aa)^*(bb)^*b$

Valid Strings:

1. "aabb" (Two as, one b)
2. "aaaaab" (Four as, one b)
3. "aabbb" (Two as, three bs)

Invalid Strings:

1. "aabbbbb" (Even number of bs)
2. "aaaabb" (Odd number of as)
3. "b" (No as present)

58. The Set of Strings Over {a, b} That Contains the Substrings aa or bb

$(a+b)^*aa(a+b)^*+(a+b)^*bb(a+b)^*$

Valid Strings:

1. "aab" (Contains aa)
2. "abba" (Contains bb)
3. "baabb" (Contains both aa and bb)

Invalid Strings:

1. "abab" (No aa or bb)
2. "ba" (No aa or bb)
3. "ababa" (No aa or bb)

59. The Set of Strings Over {a, b} That Do Not Contain the Substrings aa and bb

$(ab)^*(ab)^*aa(ab)^*+(ab)^*bb(ab)^*$

Valid Strings:

1. "ab" (Alternates)
2. "abab" (Alternates)
3. "bababa" (Alternates)

Invalid Strings:

1. "aab" (Contains aa)
2. "abba" (Contains bb)
3. "baabb" (Contains both aa and bb)

60. The Strings That Start with aa, End with bb, and Have Alternating Substrings ba in Between $a(ab)^+b$ or $aa(ba)^+bb$

Valid Strings:

1. "aabb" (Starts with aa, ends with bb)
2. "aabababb" (Has ba in between)
3. "aa" (Special case, can end immediately with bb)

Invalid Strings:

1. "abb" (Does not start with aa)
2. "aabba" (Does not end with bb)
3. "aaabaabb" (Middle part is incorrect, contains aa)

61. The Strings That Contain at Most One b and the Rest as

$a^*(b+\wedge)a^*$

Valid Strings:

1. "aaaa" (No bs)
2. "aaabaa" (One b)
3. "ba" (One b, valid)

Invalid Strings:

1. "abb" (More than one b)
2. "aabba" (More than one b)
3. "bb" (More than one b)

62. The Even-Length Strings of as and bs $(aa+bb+ab+ba)^*$

Valid Strings:

1. "ab" (Even length)
2. "abba" (Even length)
3. "aabb" (Even length)

Invalid Strings:

1. "a" (Odd length)
2. "bab" (Odd length)
3. "abbba" (Odd length)

63. The Odd-Length Strings of as and bs

$(a+b)(aa+bb+ab+ba)^*$

Valid Strings:

1. "a" (Odd length)
2. "bab" (Odd length)
3. "abbba" (Odd length)

Invalid Strings:

1. "ab" (Even length)
2. "abba" (Even length)
3. "aabb" (Even length)

64. The Set of All Strings of a's and b's That Have At Least Two Letters, Begin and End with a's, and Have Only b's in Between

ab^*a

Valid Strings:

1. "aa" (Starts and ends with a, zero b's)
2. "aba" (Starts with a, ends with a, one b in between)
3. "abbba" (Starts and ends with a, multiple b's in between)

Invalid Strings:

1. "ab" (Does not end with a)
2. "baa" (Does not start with a)
3. "bbba" (Does not start with a)

65. All Words That Begin with Either a or c and Are Followed by Some Number of b's

$(a+c)b^*$

Valid Strings:

1. "a" (Starts with a, no b's)
2. "c" (Starts with c, no b's)
3. "ab" (Starts with a, followed by b)

Invalid Strings:

1. "b" (Does not start with a or c)
2. "ba" (Starts with b, not a or c)
3. "cc" (Second character is not b)

66. The Only Words That Do Not Contain Both a and b in Them

a^+b^*

Valid Strings:

1. "" (Empty string is valid)
2. "aaaa" (Only a's)
3. "bbbb" (Only b's)

Invalid Strings:

1. "ab" (Contains both a and b)
2. "ba" (Contains both a and b)
3. "aabb" (Contains both a and b)

67. All Words That Have a Positive Number of a's Followed by Exactly One b

aa^+b

Valid Strings:

1. "ab" (One a, followed by one b)
2. "aab" (Two a's, followed by one b)
3. "aaaab" (Four a's, followed by one b)

Invalid Strings:

1. "b" (Does not start with a)
2. "ba" (Incorrect order)
3. "aa" (Missing final b)

68. All Words That Contain at Most One b and the Rest Are a's

$a^*(b+ \Lambda)a^*$

Valid Strings:

1. "aaaa" (No b, only a's)
2. "aaabaa" (One b, surrounded by a's)
3. "ba" (One b, at the start)

Invalid Strings:

1. "abb" (More than one b)
2. "aabba" (More than one b)
3. "bb" (More than one b)

69. The Language Consisting of Strings That Are Either All a's or b Followed by a Nonnegative Number of a's

a^+ba^* or $a^*(\wedge + b)$

Valid Strings:

1. "aaaa" (Only a's)
2. "b" (Starts with b, followed by zero a's)
3. "ba" (Starts with b, followed by one a)

Invalid Strings:

1. "ab" (Contains b after a, which is not allowed)
2. "bb" (Has more than one b)
3. "bba" (Has more than one b)

70. The Language That Denotes All Words With at Least Two a's

$b^*ab^*a(a+b)^*$

Valid Strings:

1. "aa" (At least two a's)
2. "baa" (Starts with b, but still has two a's)
3. "aba" (Contains at least two a's)

Invalid Strings:

1. "a" (Only one a)
2. "b" (No a's)
3. "bbb" (No a's)
4. "bab" (Only one a)

71. The Language of All Words in Which Either a Comes Before b or b Comes Before a

$(a+b)^*a(a+b)^*b(a+b)^* + (a+b)^*b(a+b)^*a(a+b)^*$

Valid Strings:

1. "ab" (a comes before b)
2. "ba" (b comes before a)
3. "aabb" (a comes before b)

Invalid Strings:

1. "aaaa" (No b)
2. "bbbb" (No a)
3. "bbaabb" (Both a and b appear, but mixed in a way that doesn't conform to one clear rule)

72. The Language of All Strings of a's and b's That Have Even Length

$(aa+ab+bb+ba)^* \text{ or } ((a+b)(a+b))^*$

Valid Strings (Even Length):

1. "" (Empty string, length = 0 )
2. "aa" (Length = 2 )
3. "ab" (Length = 2 )

Invalid Strings (Odd Length):

1. "a" (Length = 1 )
2. "aaa" (Length = 3 )
3. "abb" (Length = 3 )

73. The Set of All Strings Over {a, b, c} That Do Not Contain the Substring "ac"

$c^*(a+(bc)^*)^*$

Valid Strings (No "ac"):

1. "aaaa" (Only a's, no c)
2. "bbbb" (Only b's, no c)
3. "ccc" (Only c's)

Invalid Strings (Contain "ac"):

1. "ac" (Direct "ac" )
2. "aacb" ("ac" appears in the middle )
3. "bac" ("ac" appears at the end )

74. Language of all words that have at least one a and at least one b

$(a+b)^*a(a+b)^*b(a+b)^*+(a+b)^*b(a+b)^*a(a+b)^*$

Valid Strings:

1. "ab" (Contains at least one a and one b)
2. "ba" (Contains at least one a and one b)
3. "aabb" (Contains at least one a and one b)

Invalid Strings:

1. "aaaa" (No b)
2. "bbbb" (No a)
3. "" (Empty string does not contain both a and b)

75. All Strings with Prefix "ab"

$ab(a+b)^*$

Valid Examples:

1. "ab" → Starts with "ab" and has nothing after it (valid).
2. "abb" → Starts with "ab", followed by "b" (valid).
3. "abaa" → Starts with "ab", followed by "aa" (valid).

Invalid Examples:

1. "ba" → Starts with "ba", not "ab" (invalid).
2. "babab" → Starts with "b", not "ab" (invalid).
3. "aab" → Starts with "aa", not "ab" (invalid).

76. All Strings with Postfix "ab"

$(a+b)^*ab^*$

Valid Examples:

1. "ab" → Ends with "ab" (valid).
2. "aab" → Ends with "ab" (valid).
3. "bab" → Ends with "ab" (valid).

Invalid Examples:

1. "aba" → Ends with "a", not "ab" (invalid).
2. "bbba" → Ends with "a", not "ab" (invalid).
3. "aaa" → Ends with "a", not "ab" (invalid).

77. All Strings with Prefix "ba" or "ab"

$ba(a+b)^*+ab(a+b)^*$

Valid Examples:

1. "ba" → Starts with "ba" (valid).
2. "ab" → Starts with "ab" (valid).
3. "baa" → Starts with "ba", followed by "a" (valid).

Invalid Examples:

1. "bb" → Starts with "bb", not "ba" or "ab" (invalid).
2. "aab" → Starts with "aa", not "ba" or "ab" (invalid).
3. "b" → Does not start with "ba" or "ab" (invalid).

78. All Strings with Postfix "ba" or "ab"

$(a+b)^*ba+(a+b)^*ab$

 Valid Examples:

1. "ba" → Ends with "ba" (valid).
2. "ab" → Ends with "ab" (valid).
3. "aabba" → Ends with "ba" (valid).

Invalid Examples:

1. "aaa" → Ends with "a", not "ba" or "ab" (invalid).
2. "bbaa" → Ends with "aa", not "ba" or "ab" (invalid).
3. "bbb" → Ends with "b", not "ba" or "ab" (invalid).

79. All Strings with Prefix "ab" and Postfix "ba"

$ab(a+b)^*ba$

Valid Examples:

1. "abba" → Starts with "ab", ends with "ba" (valid).
2. "ababa" → Starts with "ab", ends with "ba" (valid).
3. "abbba" → Starts with "ab", ends with "ba" (valid).

Invalid Examples:

1. "baab" → Starts with "ba", not "ab" (invalid).
2. "abab" → Ends with "ab", not "ba" (invalid).
3. "bbba" → Does not start with "ab" (invalid).

80. All Strings with Prefix "ba" and Postfix "ab"

$ba(a+b)^*ab$

Valid Examples:

1. "baab" → Starts with "ba", ends with "ab" (valid).
2. "babab" → Starts with "ba", ends with "ab" (valid).
3. "baaaab" → Starts with "ba", ends with "ab" (valid).

Invalid Examples:

1. "abba" → Starts with "ab", not "ba" (invalid).
2. "abab" → Starts with "ab", not "ba" (invalid).
3. "aaa" → Does not start with "ba" or end with "ab" (invalid).

81. All Strings That Have Two Consecutive A's

$(a+b)^*aa(a+b)^*$

Valid strings:

1. "aa" → Contains "aa" (valid).
2. "aabaa" → Contains "aa" twice (valid).
3. "abbaaa" → Contains "aa" at the end (valid).

Invalid strings:

1. "b" → No "aa" present (invalid).
2. "abab" → No "aa" present (invalid).
3. "bbba" → No "aa" present (invalid).

82. All Strings Except Those With Two Consecutive A's

$(b^*ab)^*b^+(b^*ab)^*b^*a$

Valid strings:

1. "b" → Only "b" (valid).
2. "bab" → No consecutive "aa" (valid).
3. "babab" → No consecutive "aa" (valid).

Invalid Examples:

1. "aa" → Contains "aa" (invalid).
2. "aaba" → Contains "aa" (invalid).
3. "bbaa" → Contains "aa" (invalid).

83. All Strings with an Even Number of A's

$(b^*ab^*ab^*)^*$

Valid strings:

1. "" (empty string) → Zero as (even).
2. "bb" → Zero as (even).
3. "bbaabb" → Four as (even).

Invalid strings:

1. "a" → One a (odd, invalid).
2. "aba" → Three as (odd, invalid).
3. "aaa" → Three as (odd, invalid).

84. Language of All Even-Length Strings of Alternating A's and B's

$(ab)^* + (ba)^*$

Valid strings:

1. "" (empty string) $\rightarrow$ Even length (zero).
2. "ab" $\rightarrow$ Alternates, length = 2 (even).
3. "baba" $\rightarrow$ Alternates, length = 4 (even).

Invalid strings:

1. "a"/"b" $\rightarrow$ Length = 1 (odd, invalid).
2. "abb" $\rightarrow$ "bb" breaks the alternating pattern (invalid).
3. "baa" $\rightarrow$ "aa" breaks the alternating pattern (invalid).

85. The Set of strings Over {a, b} That End in Three Consecutive B's

$(a+b)^*bbb$

Valid strings:

1. "bbb" $\rightarrow$ Ends with "bbb" (valid).
2. "aabb" $\rightarrow$ Ends with "bbb" (valid).
3. "babbbb" $\rightarrow$ Ends with "bbb" (valid).

Invalid strings:

1. "bb" $\rightarrow$ Does not end in **three** "b"s (invalid).
2. "baab" $\rightarrow$ Ends with "b", not "bbb" (invalid).
3. "aaaa" $\rightarrow$ Ends with "a", not "bbb" (invalid).

86. The Set of Strings That Have at Least One B

$a^*b(a+b)^*$

Valid strings:

1. "b" $\rightarrow$ Contains "b" (valid).
2. "ab" $\rightarrow$ Contains "b" (valid).
3. "aaaaab" $\rightarrow$ Contains "b" (valid).

Invalid strings:

1. "" $\rightarrow$ No "b" (invalid).
2. "aaa" $\rightarrow$ No "b" (invalid).

3. "aaaaa" $\rightarrow$ No "b" (invalid).

87. The Language LLL of Strings of Odd Length Over $\Sigma=\{a\}$

$a(aa)^*$

Valid strings:

1. "a" $\rightarrow$ Length = 1 (odd).
2. "aaa" $\rightarrow$ Length = 3 (odd).
3. "aaaaa" $\rightarrow$ Length = 5 (odd).

Invalid strings:

1. "" $\rightarrow$ Length = 0 (even).
2. "aa" $\rightarrow$ Length = 2 (even).
3. "aaaa" $\rightarrow$ Length = 4 (even).

88. The Language LLL of Strings That Do Not Start with "a", Over $\Sigma=\{a,b,c\}$

$(b+c)(a+b+c)^*(b+c)(a+b+c)^*(b+c)(a+b+c)^*$

Valid strings:

1. "b" $\rightarrow$ Starts with "b" (valid).
2. "c" $\rightarrow$ Starts with "c" (valid).
3. "bb" $\rightarrow$ Starts with "b" (valid).
4. "cb" $\rightarrow$ Starts with "c" (valid).

Invalid strings:

1. "a" $\rightarrow$ Starts with "a" (invalid).
2. "ab" $\rightarrow$ Starts with "a" (invalid).

89. The Language LLL of Strings of Length 2 Over $\Sigma=\{0,1,2\}$

$(0+1+2)(0+1+2)$

Valid strings:

1. "00", "01", "02"
2. "10", "11", "12"
3. "20", "21", "22"

Invalid strings:

1. "0" → Length is 1 (invalid).
2. "000" → Length is 3 (invalid).
3. "3" → "3" is not in Σ (invalid).

90. The Language LLL of Strings Ending in "0", Over $\Sigma=\{0,1\}$

$(0+1)^*0$

Valid strings:

1. "0" → Ends in "0" (valid).
2. "10" → Ends in "0" (valid).
3. "110" → Ends in "0" (valid).

Invalid strings:

1. "" → No characters (invalid).
2. "1" → Ends in "1" (invalid).
3. "011" → Ends in "1" (invalid).

91. The Language EQUALEQUAL, of Strings Where the Number of "a"s Equals the Number of "b"s, Over $\Sigma=\{a,b\}$

$(ab)^*$

Valid Examples:

1. "" → Zero "a"s and zero "b"s (valid).
2. "ab" → 1 "a", 1 "b" (valid).
3. "abba" → 2 "a"s, 2 "b"s (valid).

Invalid Examples:

1. "a" → 1 "a", 0 "b" (invalid).
2. "abb" → 1 "a", 2 "b"s (invalid).

92. The set of strings over $\{0,1\}$ that end in three consecutive 1s

$(0+1)^*111$

Valid strings:

1. **111** → Ends with 111
2. **0111** → Ends with 111
3. **1010111** → Ends with 111

Invalid strings:

1. **000** → Does not end with 111
2. **10101** → Ends with 101, not 111
3. **1110** → Ends with 0, not 111

93. The set of strings over {0,1} that have at least one 1

$0^*1(0+1)^*$

Valid strings:

1. **1** → Contains 1
2. **01** → Contains 1
3. **000100** → Contains 1

Invalid strings:

1. **0** → No 1 present
2. **0000** → No 1 present
3. **(empty string)** → No 1 present

94. The language that consists of all strings where the length of any run of consecutive 0s is a multiple of 3

$(1+000)^*$

Valid strings:

1. **1** → No 0s present
2. **000** → A single group of three 0s
3. **10001000** → Groups of 000, separated by 1s

Invalid strings:

1. **00** → Run of 0s is not a multiple of 3
2. **1000** → Contains 1000 (group of 4 0s, which is not allowed)
3. **000100** → The last group has only 2 0s, violating the rule

95. The language of all strings that end in 1101

$(1+0)^*1101$

Valid strings:

1. **1101** → Ends with 1101

2. **01101** → Ends with 1101
3. **101101** → Ends with 1101

Invalid strings:

1. **101** → Does not end with 1101
2. **111011** → Ends with 011, not 1101
3. **001100** → Ends with 00, not 1101

96. Language that defines all even-length strings of alternating 0s and 1s

$(01)^* + (10)^*$

Valid strings:

1. **01** → Alternating and even-length
2. **1010** → Alternating and even-length
3. **010101** → Alternating and even-length

Invalid strings:

1. **0** → Odd-length
2. **1100** → Not strictly alternating
3. **010** → Odd-length, does not follow the pattern strictly

97. The language of binary strings that contain the substring 1011

$(0+1)^*1011(0+1)^*$

Valid strings:

1. **1011** → Directly contains 1011
2. **0010110** → Contains 1011 in the middle
3. **110101101** → Contains 1011 in the middle

Invalid strings:

1. **1001** → Does not contain 1011
2. **110110** → Does not contain 1011
3. **000000** → No 1s, so cannot contain 1011

98. The language of binary strings where every run of consecutive 1s has even length

Regular Expression:

$(0+11)^*$

Valid strings:

1. **00** → Only 0s, no isolated 1s
2. **11** → A single valid pair of 1s
3. **0011001100** → Every 1 appears in pairs

Invalid strings:

1. **1** → Single 1 (not paired)
2. **101** → Contains an isolated 1
3. **011101** → 111 contains an odd-length run of 1s

99. The language of all binary strings that do not contain the substring 1011

$$((0^*11^*00)^*0^*11^*0(10)^*0)^*(0^*+11^*+11^*0(10)^*+11^*0(10)^*1)$$

Valid strings (do not contain 1011):

1. **000** → No 1s at all
2. **110110** → No 1011 pattern
3. **1010** → Contains 101, but **not** 1011

Invalid strings (contain 1011):

1. **1011** → Directly contains 1011
2. **110101101** → Contains 1011 in the middle
3. **0010110** → Contains 1011 in the middle

100. The set of all strings over {0,1} that do not have the substring 111

$$(0^*+(((0^*(1+(11)))((00^*)(1+(11))))^*0^*))$$

Valid strings (do not contain 111):

1. **0** → No 1s, so no 111
2. **110110** → No consecutive three 1s
3. **10101** → No 111 present

Invalid strings (contain 111):

1. **111** → Directly contains 111
2. **101110** → Contains 111 inside
3. **0111011** → Contains 111

101. Language of all even-length strings of alternating 0s and 1s

$$(01)^*+(10)^*$$

Valid strings:

1. **01** → Alternating and even-length.
2. **1010** → Alternating and even-length.
3. **010101** → Alternating and even-length.

Invalid strings:

1. **0** → Odd-length.
2. **1100** → Not alternating.
3. **010** → Odd-length, breaks the pattern.

102. Language of all strings of 0s and 1s that have odd length

$(0+1)(00+01+10+11)^*$

Valid strings:

1. **0** → Single character (odd length).
2. **101** → Length 3 (odd), follows pattern.
3. **01101** → Length 5 (odd), follows pattern.

Invalid strings:

1. **00** → Length is even.
2. **1010** → Length is even.
3. **011011** → Length is even.

103. Set of all strings with any number of 0s followed by any number of 1s

0^*1^*

Valid strings:

1. **000** → Only 0s (valid).
2. **1111** → Only 1s (valid).
3. **0001111** → 0s followed by 1s.

Invalid strings:

1. **101** → 0s and 1s are not separated.
2. **01101** → A 0 appears after a 1.
3. **11000** → 1s appear before 0s.

104. All strings that have two consecutive 0s

$(0+1)^*00(0+1)^*$

Valid strings:

1. **00** → Contains 00.
2. **1001** → Contains 00 in the middle.
3. **110010** → Contains 00 in the middle.

Invalid strings:

1. **1** → No 00.
2. **101** → No 00.
3. **010101** → No 00.

105. All strings except those with two consecutive 0s

$(1^*01)^*1^*+(1^*01)^*1^*0$

Valid strings (do not contain 00):

1. **1** → No 00.
2. **101** → Alternates, no 00.
3. **010101** → Alternates, no 00.

Invalid strings (contain 00):

1. **00** → Directly contains 00.
2. **10001** → Contains 00 inside.
3. **110011** → Contains 00 inside.

106. All strings with an even number of 0s

$(1^*01^*01^*)^*$

Valid strings (contain an even number of 0s):

1. **11** → No 0s (even count).
2. **0101** → Two 0s, satisfies the condition.
3. **110011** → Two 0s together (even count).

Invalid strings (contain an odd number of 0s):

1. **0** → Single 0.
2. **100** → Three 0s (odd count).
3. **01010** → Three 0s (odd count).

107. All strings of length 2

$(00+01+10+11)^*$

Valid strings (length = 2):

1. **00**
2. **01**
3. **10**

Invalid strings (length $\neq$ 2):

1. **0** $\rightarrow$ Too short.
2. **000** $\rightarrow$ Too long.
3. **1101** $\rightarrow$ Too long.

108. Language of all strings that end in 1 and don't contain 00

$(1+01)^*$

Valid strings (end in 1 and no 00 inside):

1. **1** $\rightarrow$ Ends in 1, no 00.
2. **101** $\rightarrow$ Ends in 1, no 00.
3. **1101** $\rightarrow$ Ends in 1, no 00.

Invalid strings:

1. **00** $\rightarrow$ Contains 00.
2. **1001** $\rightarrow$ Ends in 1, but contains 00.
3. **11100** $\rightarrow$ Ends in 0, not 1.

109. All strings except those with two consecutive 0s

$(1*01)^*1^*+(1*01)^*1*0$

Valid strings (do not contain 00):

1. **1** $\rightarrow$ No 00.
2. **101** $\rightarrow$ Alternates, no 00.
3. **010101** $\rightarrow$ Alternates, no 00.

Invalid strings (contain 00):

1. **00** $\rightarrow$ Directly contains 00.
2. **10001** $\rightarrow$ Contains 00 inside.

3. **110011** → Contains 00 inside.

111. Language of all even-length strings of alternating 0s and 1s

$(01)^* + (10)^*$

Valid strings:

1. **01** → Alternating and even-length.
2. **1010** → Alternating and even-length.
3. **010101** → Alternating and even-length.

Invalid strings:

1. **0** → Odd-length.
2. **1100** → Not alternating.
3. **010** → Odd-length, breaks the pattern.

112. The set of strings over $\{0,1\}$ that end in 3 consecutive 1s

$(0+1)^*111$

Valid strings:

1. **111** → Ends with 111.
2. **0111** → Ends with 111.
3. **1010111** → Ends with 111.

Invalid strings:

1. **11** → Does not end with 111.
2. **1101** → Ends with 101, not 111.
3. **0000** → Does not contain 111.

113. The set of strings that have at least one 1

$0^*1(0+1)^*$

Valid strings:

1. **1** → Contains at least one 1.
2. **100** → Contains at least one 1.
3. **0010** → Contains at least one 1.

Invalid strings:

1. **0** → No 1 present.
2. **000** → No 1 present.
3. **(empty string "")** → No 1 present.

114. All strings without substring 001

$(1+01)^*(0+1)$

Valid strings (do not contain 001):

1. **1** → No 001.
2. **101** → No 001.
3. **1101** → No 001.

Invalid strings (contain 001):

1. **001** → Directly contains 001.
2. **10010** → Contains 001.
3. **1100101** → Contains 001.

115. Regular Expression for Strings with an Even Number of 1s (Including Zero 1s)

$(0+10)^*11(0+1)^*$

Valid Strings:

1. **ϵ (empty string)** → Zero 1s (even count).
2. **00** → Zero 1s (even count).
3. **1010** → Two 1s (even count).
4. **010101** → Four 1s (even count).

Invalid Strings:

1. **1** → One 1 (odd count).
2. **100** → One 1 (odd count).
3. **101** → Three 1s (odd count).

116. Language of All Strings with Even Length

$(00+01+10+11)^*$

Valid Strings (Even-Length):

1. **ϵ (empty string)** → Zero length (even).

2. **00** $\rightarrow$ Length = 2.
3. **1010** $\rightarrow$ Length = 4.
4. **110011** $\rightarrow$ Length = 6.

Invalid Strings (Odd-Length):

1. **1** $\rightarrow$ Length = 1.
2. **101** $\rightarrow$ Length = 3.
3. **11010** $\rightarrow$ Length = 5.

117. the language of all string of 0's and 1s that have odd length

$(0+1)(00+01+10+11)^*$

Valid Strings (Odd Length):

1. **0** $\rightarrow$ Length = 1
2. **000** $\rightarrow$ Length = 3 (0 + 00)
3. **101** $\rightarrow$ Length = 3 (1 + 01)
4. **11001** $\rightarrow$ Length = 5 (1 + 10 + 01)

Invalid Strings (Even Length):

1. **00** $\rightarrow$ Length = 2
2. **11** $\rightarrow$ Length = 2
3. **1001** $\rightarrow$ Length = 4

118. the language of all string of 0's and 1s that have odd length

$(00+01+10+11)^*(0+1)$

Valid Strings (Odd Length):

1. **0** $\rightarrow$ Length = 1
2. **001** $\rightarrow$ Length = 3 (00 + 1)
3. **11011** $\rightarrow$ Length = 5 (11 + 01 + 1)

Invalid Strings (Even Length):

1. **00** $\rightarrow$ Length = 2
2. **11** $\rightarrow$ Length = 2
3. **0100** $\rightarrow$ Length = 4

119. Language of all strings not containing the substring 000

$(1+0)^*10^*+(1+0)^*0^*1+(1+0)^*$

Valid Strings:

1. "1" - Valid because it doesn't contain 000.
2. "10" - Valid because it doesn't contain 000.
3. "00101" - Valid because it doesn't contain 000.

Invalid Strings:

1. "000" - Invalid because it contains 000.
2. "10001" - Invalid because it contains 000.
3. "0001" - Invalid because it contains 000.

120. Language of all strings not containing the substring 110

$$0^*1(1+0)^*+10^*(1+0)^*+(1+0)^*$$

Valid Strings:

1. "0" - Valid because it doesn't contain 110.
2. "101" - Valid because it doesn't contain 110.
3. "0101010" - Valid because it doesn't contain 110.

Invalid Strings:

1. "110" - Invalid because it contains 110.
2. "101101" - Invalid because it contains 110.
3. "1110" - Invalid because it contains 110.

121. Language of all strings containing both 101 & 010 as substrings

$$1^*(1+0)^*1^*0^*$$

Valid Strings:

1. "101010" - Valid because it contains both 101 and 010.
2. "110100101" - Valid because it contains both 101 and 010.
3. "01010101" - Valid because it contains both 101 and 010.

Invalid Strings:

1. "101" - Invalid because it only contains 101.
2. "010" - Invalid because it only contains 010.
3. "1101" - Invalid because it doesn't contain both 101 and 010.

122. Language of all strings where both the number of 0s and 1s are even

$$(11+00+(10+01)(11+00)*(10+01))^*$$

Valid Strings:

1. "1100" - Valid because both 0s and 1s appear an even number of times.
2. "1010" - Valid because both 0s and 1s appear an even number of times.
3. "0011" - Valid because both 0s and 1s appear an even number of times.

Invalid Strings:

1. "110" - Invalid because 1s appear an odd number of times.
2. "1000" - Invalid because 0s appear an odd number of times.
3. "1110" - Invalid because 1s appear an odd number of times.

123. Language of all strings where both 0s and 1s are odd

$(11+00+(10+01)(11+00)*(10+01))*(1+0)$

Valid Strings:

1. "110" - Valid because both 0s and 1s appear an odd number of times.
2. "10001" - Valid because both 0s and 1s appear an odd number of times.
3. "11101" - Valid because both 0s and 1s appear an odd number of times.

Invalid Strings:

1. "1100" - Invalid because 0s appear an even number of times.
2. "1010" - Invalid because both 0s and 1s appear an even number of times.
3. "0011" - Invalid because both 0s and 1s appear an even number of times.

124. All words that contain exactly two 1s or three 1s (not more)

$11(1+0)*0^*+0*1(1+0)^*$

Valid Strings:

1. "1100" - Valid because it contains exactly two 1s.
2. "1010" - Valid because it contains exactly two 1s.
3. "111" - Valid because it contains exactly three 1s.

Invalid Strings:

1. "1111" - Invalid because it contains more than three 1s.
2. "10001" - Invalid because it contains more than three 1s.
3. "11011" - Invalid because it contains more than three 1s.

125. All strings that have exactly one double letter (00 or 11)

$(1*01*01^*)+(0*10*10^*)$

Valid Strings:

1. "1001" - Valid because it contains exactly one double letter "00".
2. "1100" - Valid because it contains exactly one double letter "11".
3. "0011" - Valid because it contains exactly one double letter "00".

Invalid Strings:

1. "000" - Invalid because it contains more than one double letter "00".
2. "111" - Invalid because it contains more than one double letter "11".
3. "110011" - Invalid because it contains more than one double letter "11".

126. Strings where 1 is tripled or 0 is tripled, but not both

$(1*01*01*01*)+(0*10*10*10*)$

Valid Strings:

1. "11101" - Valid because it contains tripled "1"s.
2. "00010" - Valid because it contains tripled "0"s.
3. "11100" - Valid because it contains tripled "1"s.

Invalid Strings:

1. "111000" - Invalid because it contains both tripled "1"s and "0"s.
2. "000111" - Invalid because it contains both tripled "0"s and "1"s.
3. "110000" - Invalid because it contains both tripled "1"s and "0"s.

127. Strings where the number of 1s is divisible by 3

$((0*10*10*10*))^*$

Valid Strings:

1. "111" - Valid because the number of 1s is divisible by 3.
2. "110110" - Valid because the number of 1s is divisible by 3.
3. "111000" - Valid because the number of 1s is divisible by 3.

Invalid Strings:

1. "101" - Invalid because the number of 1s is not divisible by 3.
2. "1111" - Invalid because the number of 1s is not divisible by 3.
3. "1101" - Invalid because the number of 1s is not divisible by 3.

128. Strings where 0s appear in clumps of odd numbers

$1*(00+01)*0(1)^*$

Valid Strings:

1. "110" - Valid because 0s appear in clumps of odd numbers.
2. "1101100010" - Valid because 0s appear in clumps of odd numbers.
3. "1001" - Valid because 0s appear in clumps of odd numbers.

Invalid Strings:

1. "110011" - Invalid because 0s appear in clumps of even numbers.
2. "10000" - Invalid because 0s appear in clumps of even numbers.
3. "110000" - Invalid because 0s appear in clumps of even numbers.